TVA TENNESSEE VALLEY AUTHORITY

Post Office Box 2000, Decatur, Alabama 35609-2000

January 29, 2024

10 CFR 50.73
10 CFR 50.4(a)

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Unit 1
Renewed Facility Operating License No. DPR-33
NRC Docket No. 50-259

Subject: **Licensee Event Report 50-259/2023-003-00 – Standby Liquid Control Inoperable due to Demineralized Water In-leakage**

The enclosed Licensee Event Report provides details of an inoperability of the Standby Liquid Control System. The Tennessee Valley Authority (TVA) is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(v)(C) & (D), as any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material; or mitigate the consequences of an accident.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact David Renn, Site Licensing Manager, at (256) 729-2636.

Respectfully,

Manu Sivaraman
BFN Site Vice President

Enclosure: Licensee Event Report 50-259/2023-003-00 – Standby Liquid Control Inoperable due to Demineralized Water In-leakage

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

NRC FORM 366 (10-01-2023)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 03/31/2024

LICENSEE EVENT REPORT (LER)

1. Facility Name		2. Docket Number	3. Page
Browns Ferry Nuclear Plant, Unit 1	☒ 050 ☐ 052	00259	1 OF 7

4. Title
Standby Liquid Control Inoperable due to Demineralized Water In-leakage

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
01	18	2023	2023	- 003 -	00	01	29	2024	N/A	05000 N/A
									N/A	05000 N/A

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Denzel Housley, Licensing Engineer	256-729-7643

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

4. Supplemental Report Expected)		15. Expected Submission Date	Month	Day Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date		N/A	N/A

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

A past operability evaluation determined that for a period of time from January 18 to January 19, 2023, the Standby Liquid Control (SLC) system tank sodium pentaborate (SPB) concentration dropped below the requirements of Technical Specification (TS) Surveillance Requirement 3.1.7.3 which requires the SPB solution to be greater than or equal to 8.0% by weight. The cause of this event was leakage past valve 1-SHV-002-1208, “SLC Demin Water Supply Shutoff Valve,” that allowed demineralized water to dilute the SPB solution in the SLC tank.

The past operability evaluation performed by engineering concluded that although SPB concentration dropped below the TS minimum requirements, sufficient SPB solution was available to provide the required pH buffering function; therefore, the safety function of the SLC system was maintained.

An engineering change to physically separate the SLC tank from the demineralized water system has been implemented to prevent leakage into the SLC tank.

NARRATIVE

I. Plant Operating Conditions before the Event

At the time of this event, Browns Ferry Nuclear Plant (BFN) Unit 1 was in Mode 1 at approximately 100 percent power.

II. Description of Event

A. Event Summary

On January 19, 2023, engineering noted that the Standby Liquid Control (SLC) system [BR] tank level had increased greater than 1 inch. The SLC tank level was validated by Chemistry per dipstick method and discovered to have risen approximately 2 inches. The cause of the SLC tank level increase was determined to be demineralized water [KC] leaking past valve 1-SHV-002-1208, “SLC Demin Water Supply Shutoff Valve.” The valve was adjusted to stop the leakage of demineralized water into the SLC tank. Chemistry added sodium pentaborate (SPB) to the tank and then analyzed the SLC tank contents to ensure proper limits were met.

Following a similar event in November 2023, the operability of the SLC system between the start of the water intrusion and the addition of SPB by Chemistry was questioned for both events. Past operability evaluations were requested for these events on November 14, 2023.

Past operability evaluations were performed by engineering on the two events. On November 30, 2023, it was determined that on January 18, 2023, the SLC tank SPB concentration dropped below the requirements of Technical Specification (TS) Surveillance Requirement (SR) 3.1.7.3 which requires the SPB solution to be greater than or equal to 8.0% by weight. Failure to meet TS SR 3.1.7.3 resulted in the SLC system being inoperable. Because the SLC tank is common to both SLC subsystems, the Tennessee Valley Authority (TVA) is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(v)(C) & (D), as any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to control the release of radioactive material; or mitigate the consequences of an accident.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

NARRATIVE

C. Dates and approximate times of occurrences

DATE AND APPROXIMATE TIMES (times are Central Standard Time)	**OCCURRENCE**
January 18, 2023, at 10:35	SLC tank level starts to increase.
January 18, 2023, at approximately 20:32	Based on the engineering evaluation, the SLC tank SPB concentration dropped below the TS SR requirement of 8.0%.
January 19, 2023, at 14:05	Operations declared the SLC system operable following SPB addition to the SLC tank. Testing of the SLC tank testing showed SPB concentration above 8.0%. Time duration with SPB concentration less than the TS SR requirement was approximately 17.5 hours. Minimum SPB concentration during the event was 7.96%.
November 14, 2023	Condition Reports (CR) 1892408 and 1892410 were initiated to request the performance of past operability evaluations.
November 30, 2023	Past operability evaluation was completed by engineering which showed that January event resulted in not meeting TS SR requirement for minimum SPB concentration

D. Manufacturer and model number of each component that failed during the event

No components failed during this event.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

In-leakage of demineralized water to the SLC tank through 1-SHV-002-1208 was discovered by engineering observation.

NARRATIVE

G. The failure mode, mechanism, and effect of each failed component

No components failed during this event.

H. Operator actions

Operators took actions to halt the leakage of demineralized water to the SLC tank.

I. Automatically and manually initiated safety system responses

There were no automatic or manual safety system responses associated with this event.

III. Cause of the event

A. Cause of each component or system failure or personnel error

The cause of this event was leakage past valve 1-SHV-002-1208 that allowed demineralized water to dilute the SPB solution in the SLC tank.

B. Cause(s) and circumstances for each human performance related root cause

No human performance related root causes were identified.

IV. Analysis of the event

At the time of this event, the possibility of in-leakage of demineralized water through valve 1-SHV-002-1208 after manipulation was recognized. 1-SHV-002-1208 is a one-of-a-kind valve at BFN; it is a ratchet valve which only turns in one direction, and has no backstop. Actions had been taken for periodic review of SLC tank levels by engineering and to generate an engineering change to physically isolate the demineralized water supply from the SLC tank by means of a blind flange. At the time of the event, Engineering Change Package (ECP) BFN-23-029-01 had been issued for this modification on Unit 1 but had not yet been implemented.

Operator response to SLC tank dilution events were performed under TS SR 3.1.7.3 Frequency requirement which states: “once within 24 after water or boron is added to solution.” Accordingly, actions were taken to restore the SLC tank SPB concentration within 24 hours and not to assess the resultant SPB solution due to the water addition. When questioned about this response, actions were initiated to perform past operability evaluations.

NARRATIVE

V. Assessment of Safety Consequences

The SLC System is credited in the radiological dose analysis for a loss of coolant accident (LOCA) to provide a buffering agent (SPB solution) to the suppression pool water. In determining the fission products available for release from the containment, the analysis credits fission product scrubbing in the suppression pool. Once deposited in the suppression pool water, the iodine will remain in solution as long as the suppression pool pH remains above 7.0. In calculating the post-LOCA suppression pool pH, credit must be taken for buffering of the suppression pool water by the SPB injected by the SLC System to counterbalance the production of acids in the containment post-LOCA. The volume of SPB required by the post-LOCA suppression pool pH calculated is based upon the available boron and is not affected by the boron-10 enrichment.

The SLC System is also designed to provide the capability of bringing the reactor, at any time in a fuel cycle, from full power and minimum control rod inventory to a subcritical condition with the reactor in the most reactive, xenon free state without taking credit for control rod movement. This function, however, is not credited as a safety function.

The past operability evaluation of this event reviewed the available SPB solution during this event and determined that sufficient SPB solution was available to fulfill the buffering requirement for the LOCA. Additionally, the SLC system was operable within the TS required completion times of TS 3.1.7, “Standby Liquid Control (SLC) System.” With two SLC subsystems inoperable, TS 3.1.7 Condition B requires one SLC subsystem to be restored to Operable status within 8 hours. If Condition B is not met, then TS 3.1.7, Condition C.1 requires the plant be in Mode 3 in 12 hours. In this event, the SLC system was inoperable for approximately 17.5 hours which is less than the total TS time of 20 hours to be in Mode 3.

Based on the above, during the time period that the SLC system was inoperable, sufficient SPB solution was available to provide the required pH buffering function to protect the health and safety of the public. There was no significant reduction to the health and safety of the public or plant personnel for this event. This event was resolved within the TS-required completion time.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

Although both subsystems of SLC were inoperable due to TS SR requirements for the common SLC tank, the past operability evaluation performed by engineering concluded that sufficient SPB solution was available to provide the required pH buffering function; therefore, the SLC system could still perform its safety function during this event.

NARRATIVE

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shutdown.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

As assessed by the past operability evaluation, the SPB solution concentration was less than required by TS SR 3.1.7.3 for approximately 17.5 hours.

VI. Corrective Actions

Corrective Actions were managed by the TVA’s corrective action program under CR 1892410.

A. Immediate Corrective Actions

The in-leakage of demineralized water to the SLC tank was stopped, SPB solution was added to the SLC tank, and analysis of the SLC tank SPB solution was performed to ensure TS requirements were met.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

ECP BFN-23-029-01, which physically separates the Unit 1 SLC tank from the demineralized water system, has been implemented to prevent leakage into the SLC tank.

Additionally, as an enhancement, 0-TI-18, “Sodium Pentaborate Concentration and Level Adjustment of the Standby Liquid Control System,” has been revised to require a SLC sample be obtained prior to a chemical addition to a SLC tank of unknown concentration.

VII. Previous Similar Events at the Same Site

A search of LERs from BFN, Units 1, 2, and 3 over the last five years identified no similar events.

VIII. Additional Information

There is no additional information.

NARRATIVE

IX. Commitments

There are no new commitments.

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].